Mock Test > yehdan@msn.com

Full Name:	Jianping Ye
Email:	yehdan@msn.com
Test Name:	**Mock Test**
Taken On:	9 Aug 2025 21:59:27 IST
Time Taken:	11 min 43 sec/ 25 min
Invited by:	Ankush
Invited on:	9 Aug 2025 21:59:19 IST
Skills Score:	
Tags Score:	Algorithms 75/75 Core CS 75/75 Medium 75/75 Search 75/75 problem-solving 75/75

100%

75/75

scored in **Mock Test** in 11 min 43 sec on 9 Aug 2025 21:59:27 IST

Recruiter/Team Comments:

No Comments.

	Question Description	Time Taken	Score	Status
Q1	Pairs > **Coding**	11 min 26 sec	75/ 75	✓

QUESTION 1

Correct Answer

Score 75

Pairs > Coding Search Algorithms Medium problem-solving Core CS

QUESTION DESCRIPTION

Given an array of integers and a target value, determine the number of pairs of array elements that have a difference equal to the target value.

Example

$k = 1$

$arr = [1, 2, 3, 4]$

There are three values that differ by $k = 1$: $2 - 1 = 1$, $3 - 2 = 1$, and $4 - 3 = 1$. Return 3.

Function Description

Complete the *pairs* function below.

pairs has the following parameter(s):

- *int k:* an integer, the target difference
- *int arr[n]:* an array of integers

Returns

- *int:* the number of pairs that satisfy the criterion

Input Format

The first line contains two space-separated integers n and k, the size of arr and the target value.
The second line contains n space-separated integers of the array arr.

Constraints

- $2 \leq n \leq 10^5$
- $0 < k < 10^9$
- $0 < arr[i] < 2^{31} - 1$
- each integer $arr[i]$ will be unique

Sample Input

```
STDIN       Function
-----       --------
5 2         arr[] size n = 5, k =2
1 5 3 4 2   arr = [1, 5, 3, 4, 2]
```

Sample Output

```
3
```

Explanation

There are 3 pairs of integers in the set with a difference of 2: [5,3], [4,2] and [3,1]. .

CANDIDATE ANSWER

Language used: **C++14**

```

std::vector<int> find_identical_elements(const std::vector<int>& list1, const
std::vector<int>& list2) {
    std::vector<int> common_elements;
    auto it1 = list1.begin();
    auto it2 = list2.begin();

    while (it1 != list1.end() && it2 != list2.end()) {
        if (*it1 == *it2) {
            common_elements.push_back(*it1);
            ++it1;
            ++it2;
        } else if (*it1 < *it2) {
            ++it1;
        } else { // *it1 > *it2
            ++it2;
        }
    }
    return common_elements;
}

/*
 * Complete the 'pairs' function below.
 *
 * The function is expected to return an INTEGER.
 * The function accepts following parameters:
 *  1. INTEGER k
 *  2. INTEGER_ARRAY arr
 */
```

```
int pairs(int k, vector<int> arr) {
    // sort first
    std::sort(arr.begin(), arr.end());

    // + k
    vector<int> plusk;
    for(auto a : arr) plusk.push_back(a + k);

    // find identical elements

    auto res = find_identical_elements(arr, plusk);

    return res.size();
}
```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 1	Easy	Hidden case	Success	5	0.0104 sec	8.38 KB
Testcase 2	Easy	Hidden case	Success	5	0.0087 sec	8.75 KB
Testcase 3	Easy	Hidden case	Success	5	0.0091 sec	8.75 KB
Testcase 4	Easy	Hidden case	Success	5	0.009 sec	8.75 KB
Testcase 5	Easy	Hidden case	Success	5	0.0087 sec	8.63 KB
Testcase 6	Easy	Hidden case	Success	5	0.0119 sec	8.91 KB
Testcase 7	Easy	Hidden case	Success	5	0.0139 sec	8.84 KB
Testcase 8	Easy	Hidden case	Success	5	0.0104 sec	8.5 KB
Testcase 9	Easy	Hidden case	Success	5	0.0112 sec	8.81 KB
Testcase 10	Easy	Hidden case	Success	5	0.0124 sec	8.89 KB
Testcase 11	Easy	Hidden case	Success	5	0.0423 sec	14.2 KB
Testcase 12	Easy	Hidden case	Success	5	0.0388 sec	14.3 KB
Testcase 13	Easy	Hidden case	Success	5	0.0448 sec	14.3 KB
Testcase 14	Easy	Hidden case	Success	5	0.0427 sec	14.3 KB
Testcase 15	Easy	Hidden case	Success	5	0.0433 sec	14 KB
Testcase 16	Easy	Sample case	Success	0	0.0086 sec	8.5 KB
Testcase 17	Easy	Sample case	Success	0	0.0098 sec	8.38 KB
Testcase 18	Easy	Sample case	Success	0	0.0098 sec	8.38 KB

No Comments